



Qualification Pack

Technical Supervisor - Computer Aided Engineering

QP Code: MSME/CSC/Q3301

Version: 1.0

NSQF Level: 5

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Qualification Pack

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Qualification Pack

MSME/CSC/Q3301: Technical Supervisor - Computer Aided Engineering

Brief Job Description

The learner who attend this qualification will be competent in the following area To be expertise in Engineering Analysis Solution through CAE Application

Personal Attributes

The learner who attend this qualification will be competent in the following area To be expertise in Engineering Analysis Solution through CAE Application

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MSME/CSC/N3303: Analyses & Simulation of CAD Model using CAE Tools](#)
2. [MSME/CSC/N3302: Create & Modify Part Model using higher end CAD Software](#)
3. [MSME/CSC/N3301: INTRODUCTION TO FINITE ELEMENT ANALYSIS \(FEA\)](#)
4. [MSME/CSC/N3304: Employability skills 05](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Machine Tools
Occupation	Design analysis
Country	India
NSQF Level	5
Credits	20
Aligned to NCO/ISCO/ISIC Code	(Tool Engineer)



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Minimum Educational Qualification & Experience	B.E./B.Tech (BE/B.Tech Perusing in relevant Trade) with NA of experience OR Completed 3 year diploma after 10th with 4.5 years of experience OR Certificate-NSQF (Previous relevant Qualification of NSQF Level 4.5) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (Previous relevant Qualification of NSQF Level 4.5) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	NQR Code: QG-05-CG-02407-2024-V1-MSME
NQR Version	1.0



Qualification Pack

MSME/CSC/N3303: Analyses & Simulation of CAD Model using CAE Tools

Description

After completion of course Student should be able to Able to Analyze and understand customers need.

Scope

The scope covers the following :

- After completion of course Student should be able to Able to Analyze and understand customers need.

Elements and Performance Criteria

MSME/MCCCAE/03 Analyses & Simulation of CAD Model using CAE Tools

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the need Analysis and need of CAE Software
- PC2.** Different file format use in CAE Software
- PC3.** Export and Import of different file format in CAE
- PC4.** Define the different parameter for analysis
- PC5.** Generate the mesh file
- PC6.** Define the load and boundary parameter for the CAD model
- PC7.** Simulate the result
- PC8.** Validate the result
- PC9.** Optimize the result
- PC10.** Generate the macro file for the analysis
- PC11.** Generate the different types of graphs
- PC12.** Publish the graphs
- PC13.** Generate the report.
- PC14.** Generate the Auto Meshing & Manual Meshing.
- PC15.** Generate the Geometry and modify the geometry using Hypermesh



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/MCCCAE/03 Analyses & Simulation of CAD Model using CAE Tools</i>	-	100	-	-
PC1. Understand the need Analysis and need of CAE Software	-	-	-	-
PC2. Different file format use in CAE Software	-	-	-	-
PC3. Export and Import of different file format in CAE	-	-	-	-
PC4. Define the different parameter for analysis	-	-	-	-
PC5. Generate the mesh file	-	-	-	-
PC6. Define the load and boundary parameter for the CAD model	-	-	-	-
PC7. Simulate the result	-	-	-	-
PC8. Validate the result	-	-	-	-
PC9. Optimize the result	-	-	-	-
PC10. Generate the macro file for the analysis	-	-	-	-
PC11. Generate the different types of graphs	-	-	-	-
PC12. Publish the graphs	-	-	-	-
PC13. Generate the report.	-	-	-	-
PC14. Generate the Auto Meshing & Manual Meshing.	-	-	-	-
PC15. Generate the Geometry and modify the geometry using Hypermesh	-	-	-	-
NOS Total	-	100	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N3303
NOS Name	Analyses & Simulation of CAD Model using CAE Tools
Sector	Capital Goods
Sub-Sector	
Occupation	Design analysis
NSQF Level	5
Credits	12
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/CSC/N3302: Create & Modify Part Model using higher end CAD Software

Description

After completion of course Student should be able to Understand advance Computer aided design software (UNIGRAPHICS CAD) as compare to other CAD software.

Scope

The scope covers the following :

- After completion of course Student should be able to Understand advance Computer aided design software (UNIGRAPHICS CAD) as compare to other CAD software.

Elements and Performance Criteria

MSME/MCCCAE/02 Create & Modify Part Model using higher end CAD Software

To be competent, the user/individual on the job must be able to:

- PC1.** Describe various feature in CAD software.
- PC2.** Explain about limitation & advantages of each CAD Software
- PC3.** Explain working of CAD/CAM
- PC4.** Create and Edit 2D geometric sketches by using Unigraphics NX software.
- PC5.** Develop 3D modeling by using advanced command.
- PC6.**
 - Apply assembly constraint & develop different types of assembly design by
 - using CAD Software like Unigraphics NX
- PC7.** Create the detail drafting of the product / assembly
- PC8.** Create / import entities in 3D space as per job requirement
- PC9.** Modify entities in 3D space as per job requirement
- PC10.**
 - Create 3-D views on the screen by manipulating drawing planes and inserting
 - 3-D geometric shapes
- PC11.** Produce a model for export to the CAE Software



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/MCCCAE/02 Create & Modify Part Model using higher end CAD Software</i>	-	100	-	-
PC1. Describe various feature in CAD software.	-	-	-	-
PC2. Explain about limitation & advantages of each CAD Software	-	-	-	-
PC3. Explain working of CAD/CAM	-	-	-	-
PC4. Create and Edit 2D geometric sketches by using Unigraphics NX software.	-	-	-	-
PC5. Develop 3D modeling by using advanced command.	-	-	-	-
PC6. • Apply assembly constraint & develop different types of assembly design by • using CAD Software like Unigraphics NX	-	-	-	-
PC7. Create the detail drafting of the product / assembly	-	-	-	-
PC8. Create / import entities in 3D space as per job requirement	-	-	-	-
PC9. Modify entities in 3D space as per job requirement	-	-	-	-
PC10. • Create 3-D views on the screen by manipulating drawing planes and inserting • 3-D geometric shapes	-	-	-	-
PC11. Produce a model for export to the CAE Software	-	-	-	-
NOS Total	-	100	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N3302
NOS Name	Create & Modify Part Model using higher end CAD Software
Sector	Capital Goods
Sub-Sector	
Occupation	Design analysis
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/CSC/N3301: INTRODUCTION TO FINITE ELEMENT ANALYSIS (FEA)

Description

Analyze and understand the mechanical property of the component

Scope

The scope covers the following :

- Analyze and understand the mechanical property of the component

Elements and Performance Criteria

MSME/MCCCAE/01 INTRODUCTION TO FINITE ELEMENT ANALYSIS (FEA)

To be competent, the user/individual on the job must be able to:

- PC1.** Analyze and understand the mechanical property of the component.
- PC2.** Understand the concept & principle of computer aided Engineering (CAE).
- PC3.** Describe stress strain relationship
- PC4.** Solve static Analysis
- PC5.** Understand the axial stress on trusses and frame
- PC6.** Describe conduction, convection & radiation
- PC7.** To apply proper remote boundary conditions
- PC8.** Knowledge of post processing.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/MCCCAE/01 INTRODUCTION TO FINITE ELEMENT ANALYSIS (FEA)</i>	100	-	-	-
PC1. Analyze and understand the mechanical property of the component.	-	-	-	-
PC2. Understand the concept & principle of computer aided Engineering (CAE).	-	-	-	-
PC3. Describe stress strain relationship	-	-	-	-
PC4. Solve static Analysis	-	-	-	-
PC5. Understand the axial stress on trusses and frame	-	-	-	-
PC6. Describe conduction, convection & radiation	-	-	-	-
PC7. To apply proper remote boundary conditions	-	-	-	-
PC8. Knowledge of post processing.	-	-	-	-
NOS Total	100	-	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N3301
NOS Name	INTRODUCTION TO FINITE ELEMENT ANALYSIS (FEA)
Sector	Capital Goods
Sub-Sector	
Occupation	Design analysis
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/CSC/N3304: Employability skills 05

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century,
- digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service,
- entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Elements and Performance Criteria

MSME/ES/02 Employability skills

To be competent, the user/individual on the job must be able to:

- PC1.** Explain occupational health and Safety.
- PC2.** Explain about safety rules.
- PC3.**
 - State the name and location of people responsible for health and safety in the workplace
- PC4.**
 - Identify employability skills required for jobs in various industries. & Identify and explore learning and employability portals
- PC5.**
 - Recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC6.**
 - Follow environmentally sustainable practices. & Recognize the significance of 21st Century Skills for employment
- PC7.**
 - Practice the 21st Century Skills such as Self-Awareness, Behavior Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.**
 - Use basic English for everyday conversation in different contexts, in person and over the telephone.
- PC9.** How to Minimize the team conflicts & Explain Ethics & values?
- PC10.**
 - Read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.**
 - Write short messages, notes, letters, e-mails etc. in English & Understand the difference between job and career
- PC12.**
 - Prepare a career development plan with short- and long-term goals, based on aptitude & Discuss the main types of electronic funds transfers



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- PC13.** • Follow verbal and non-verbal communication etiquette and active listening
• techniques in various settings & work collaboratively with others in a team
- PC14.** • Communicate and behave appropriately with all genders and PwD & escalate
• any issues related to sexual harassment at workplace according to POSH Act.
- PC15.** • Select financial institutions, products and services as per requirement & carry
• out offline and online financial transactions, safely and securely.
- PC16.** • Identify common components of salary and compute income, expenses, taxes,
• investments etc & identify relevant rights and laws and use legal aids to fight
• against legal exploitation
- PC17.** • Operate digital devices and carry out basic internet operations securely and
• safely & use e- mail and social media platforms and virtual collaboration tools
• to work effectively
- PC18.** Use basic features of word processor, spreadsheets, and presentations.
- PC19.** • Identify different types of Entrepreneurship and Enterprises and assess
• opportunities for potential business through research & develop a business
• plan and a work model, considering the 4Ps of Marketing Product, Price, Place and
• Promotion.
- PC20.** • Identify sources of funding, anticipate, and mitigate any financial/ legal hurdles
• for the potential business opportunity
- PC21.** • Identify different types of customers & identify and respond to customer
• requests and needs in a professional manner.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/ES/02 Employability skills</i>	100	-	-	-
PC1. Explain occupational health and Safety.	-	-	-	-
PC2. Explain about safety rules.	-	-	-	-
PC3. • State the name and location of people responsible for health and safety in the • workplace	-	-	-	-
PC4. • Identify employability skills required for jobs in various industries. & Identify • and explore learning and employability portals	-	-	-	-
PC5. • Recognize the significance of constitutional values, including civic rights and • duties, citizenship, responsibility towards society etc. and personal values and • ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC6. • Follow environmentally sustainable practices. & Recognize the significance of • 21st Century Skills for employment	-	-	-	-
PC7. • Practice the 21st Century Skills such as Self-Awareness, Behavior Skills, time • management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for • continuous learning etc. in personal and professional life	-	-	-	-
PC8. • Use basic English for everyday conversation in different contexts, in person and • over the telephone.	-	-	-	-
PC9. How to Minimize the team conflicts & Explain Ethics & values?	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. • Read and understand routine information, notes, instructions, mails, letters • etc. written in English	-	-	-	-
PC11. • Write short messages, notes, letters, e-mails etc. in English & Understand the • difference between job and career	-	-	-	-
PC12. • Prepare a career development plan with short- and long-term goals, based on • aptitude & Discuss the main types of electronic funds transfers	-	-	-	-
PC13. • Follow verbal and non-verbal communication etiquette and active listening • techniques in various settings & work collaboratively with others in a team	-	-	-	-
PC14. • Communicate and behave appropriately with all genders and PwD & escalate • any issues related to sexual harassment at workplace according to POSH Act.	-	-	-	-
PC15. • Select financial institutions, products and services as per requirement & carry • out offline and online financial transactions, safely and securely.	-	-	-	-
PC16. • Identify common components of salary and compute income, expenses, taxes, • investments etc & identify relevant rights and laws and use legal aids to fight • against legal exploitation	-	-	-	-
PC17. • Operate digital devices and carry out basic internet operations securely and • safely & use e- mail and social media platforms and virtual collaboration tools • to work effectively	-	-	-	-
PC18. Use basic features of word processor, spreadsheets, and presentations.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. - Identify different types of Entrepreneurship and Enterprises and assess - opportunities for potential business through research & develop a business - plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion.	-	-	-	-
PC20. - Identify sources of funding, anticipate, and mitigate any financial/ legal hurdles - for the potential business opportunity	-	-	-	-
PC21. - Identify different types of customers & identify and respond to customer - requests and needs in a professional manner.	-	-	-	-
NOS Total	100	-	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/CSC/N3304
NOS Name	Employability skills 05
Sector	Capital Goods
Sub-Sector	
Occupation	Design analysis
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

As per QP

Minimum Aggregate Passing % at QP Level : 40

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MSME/CSC/N3303.Analyses & Simulation of CAD Model using CAE Tools	-	100	-	-	100	20
MSME/CSC/N3302.Create & Modify Part Model using higher end CAD Software	-	100	-	-	100	20
MSME/CSC/N3301.INTRODUCTION TO FINITE ELEMENT ANALYSIS (FEA)	100	-	-	-	100	40
MSME/CSC/N3304.Employability skills 05	100	-	-	-	100	20
Total	200	200	-	-	400	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.